Real-Time Traffic Accident Prediction and Proactive Vehicle Rerouting Using Temporal Graph Neural Networks and Knowledge Distillation

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***Abstract* — Accurate real-time traffic accident prediction enables proactive interventions that prevent incidents before occurrence; however, existing systems predominantly operate in offline batch mode and neglect spatial danger propagation across connected road topology. This paper presents a Temporal Graph Neural Network (T-GNN) ensemble for real-time accident prediction and proactive vehicle rerouting, deployed within a live SUMO microscopic traffic simulation. By the way, a two-level knowledge distillation pipeline substantially reduces t-gnn parameters with negligible recall cost, enabling inference within the simulation step budget. On a large-scale SUMO corpus, the T-GNN student achieves high AUC and recall while significantly reducing false positives over Random Forest. An MLP baseline isolates the measurable recall contribution of graph topology over topology-agnostic deep learning. Across multiple independent live simulation runs, the T-GNN+RF ensemble consistently outperforms the RF-only baseline on recall, F2-score, and prediction lead time, exceeding the operational target by a substantial margin. Cross-dataset validation on the pNEUMA real-world urban drone trajectory dataset confirms architectural generalisability across all leave-one-recording-out evaluations, with the simulation-to-real performance gap attributed to domain shift rather than model overfitting. A key finding is that naive graph propagation applied to classical ML outputs yields negligible improvement, confirming that the neural T-GNN architecture, not graph structure alone, is responsible for the observed gains.**

Index Terms* — *Traffic accident prediction, temporal graph neural network, knowledge distillation, SUMO simulation, vehicle rerouting, MLP baseline, optimizer comparison, F2 threshold.

I. Introduction

Globally, there are many injuries and significant economic loss from traffic accidents. To intervene before an accident occurs, accurate predictions of an accident's occurrence in advance are needed. These predictive models will usually help reroute vehicles from their original routes to avoid becoming involved in an accident. The few that do exist are primarily batch-mode systems, which process historical crash data instead of responding to traffic conditions (live traffic) and assisting the public during the accident.

This paper focuses on real-time, topology-aware accident prediction from a modelling perspective in a simulated live traffic environment. The authors argue that accidents are not the result of isolated predictive visual cues. Instead, dangerous road conditions propagate from one connected road segment to the next with decelerated vehicles on the first road segment from the accident causing traffic queues to develop and cause instability on the next road segment in the system. If the predictive model only uses per-edge positional or predictive visual features the predictive model will miss part of the accident's cause, which is spatial propagation through connected roadway segments.This paper makes five contributions:

1. A two-staged ensemble T-GNN is used to create danger scores that are aware of the topology of a graph at run-time using knowledge distillation to build a 1-hop GraphSAGE student together with a Random Forest student.

2. A baseline of MLP deep learning, using the same 625k corpus as the training set, is developed using optimisers (Adam, SGD+Nesterov, AdamW), showing that will not perform as well as topology-aware models as well as topological-based prediction.

3. Live SUMO simulations were deployed and evaluated using event-based metrics through 5 independent runs (153 -- 225 accident events) creating a repeatable baseline of XGBoost + SMOTE from the literature for comparison of architecture.

4. A 3-tier proactive rerouting subsystem is developed and the contribution of the each design component is quantitatively evaluated through a thorough ablation study.

5. Cross-dataset validation was performed on the real-world pNEUMA urban trajectory dataset, with statistical analysis of "leave-one-recording-out," confirming that T-GNN has outperformed RF in all 4 of the independent recordings (4/4 LORO Wins ; AUC: T-GNN=0.8510, RF=0.8410).

II. Related Work

A. Classical Machine Learning for Traffic Safety

Early attempts at predicting accidents using data have made use of logistic regression, decision trees, and naive Bayes. The AUC for these early methods was observed to be between 0.65 and 0.75 [16]. However, with the introduction of ensemble methods, AUC values increased significantly: using Random Forest models on historical accident data, the reported AUC ranged from 0.80 to 0.90 [1],[17]. Using XGBoost on loop detector data from Chicago expressways as a real-world reference benchmark, Parsa et al. [1] reported an AUC of 0.89. Further, Chen et al. [2] proposed the MSCPO-XGBoost method utilizing SMOTE for oversampling and achieved an AUC of 0.9282 with Chinese crash record data. Iranitalab and Khattak [18] conducted a comparison of six classifiers in the prediction of crash severity and found that the Random Forest classifier consistently outperformed logistic regression and SVM. All of the previous methods treat each road segment completely independent of each other and do not account for how the danger from a crash accumulates through the connected topology of roads. Using the same architecture as Chen et al. on our 625,577 accident samples provides an equivalent control/architectural baseline comparison.

B. Deep Learning — Sequential and Recurrent Models

Traffic flow sequences have a temporal dependence which can be modeled using recurrent neural networks (RNNs). In the realm of detecting traffic incidents, for example, LSTM-based models applied to loop detector time series have been shown to outperform classical machine learning algorithms for incident detection purposes [3],[19]. Abdulkareem [4] uses attention-based and bidirectional LSTM networks within the simulator (SUMO) to predict traffic congestion — the only known work from SUMO to date that was based on deep learning; addressing congestion not accidents; no graph topologies or rerouting. Yuan el al. [20] uses a hybrid approach (CNN+LSTM) to predict risks for accidents, using trajectory data and a unique approach in which convolutional features improve the temporal feature extraction process. Transformer-based architectures have recently been applied to traffic prediction [15]; however, they require very intensive large pre-trained datasets that are not accessible for the safety-critical accident prediction task, resulting in a very low performance of inference with the 1-second at step deployment. Our Phase 7 evaluation supports the use of the LSTM as outlined above. Train LSTM networks on 400K SUMO samples yields an AUC=0.562 with precision=7.3%, which substantiates the use of the graph topology approach.

C. Graph Neural Networks for Traffic

In 2017, Yu et al. first demonstrated that GNNs could be used effectively for traffic prediction with Spatio-Temporal Graph Convolutional Networks (STGCN); they achieved state-of-the-art results using METR-LA and PEMS-BAY datasets by combining ChebNet graph convolutions and gated temporal convolutions, and have since led to additional advancements in traffic forecasting speed. DCRNN was introduced by Li et al. through utilizing bidirectional diffusion convolutions (BDC) on directed road graphs as a way to model both upstream and downstream traffic propagation. Wu et al. developed Graph WaveNet, which introduced an adaptive adjacency matrix and the use of learned hidden spatial dependencies to improve forecast speeds even further. Attention-based graph convolution is used to predict accident hotspots across entire cities by utilizing historical accident data through the work of Zheng et al.. Real-time prediction of traffic speeds is being addressed by Bao et al. with a dynamic graph attention network, which utilizes current traffic conditions to adjust edge weights.

Li and Chen's work demonstrate the usefulness of combining CNNs, LSTMs, and GNNs for predicting future accidents based on vehicle trajectory data with a recall of 93.7%. RASPNet is a severity classification method that develops GNN and LSTM methods and utilizes over 2.8 million records from the US as input data; this demonstrates the graph-based method's effectiveness; however, both examples only demonstrate batch processing of historical data without any real-time simulations of traffic by incorporating vehicle rerouting based on these technologies. This work was validated within an ablation study and determined a maximum 10.5% contribution to a graph topology structure over a typical recall method (MLP vs T-GNN), providing a controlled measurement absent from prior work.

D. Knowledge Distillation for Efficient Deployment

Hinton and colleagues developed knowledge distillation [8], in which a compact "student" model is trained to produce soft probability outputs that are similar to those produced by a larger "teacher" model, allowing for inference with a real-time deployment ready model. Izadi and colleagues demonstrated how to apply knowledge distillation (KD) to spatio-temporal Graph Neural Networks (GNNs) for traffic prediction; by using KD, they were able to achieve an approximate 3x speed-up while only incurring <2% in accuracy loss. Zhang and colleagues [23] describe a new feature-based KD method for GNNs, transferring intermediate representations instead of just final output (as was done in our Phase 13 training where alpha_feat=0.4). Gou and colleagues [24] review KD methods thoroughly and divide them into response-based and feature-based prior

categories, with both currently being the most widely utilized. Our two-tier KD methods achieved an 85.5% reduction in the number of parameters (14,977 to 2,177) at a cost of <0.95% in recall, enabling real-time inference to occur within a 1 second constraint of the SUMO simulation step.

E. Vehicle Rerouting and Traffic Management

There are also many studies evaluating the proactive rerouting of vehicles based on predicted incidents (also called dynamic rerouting) using variable message signs and navigation systems. Zhu et al.[25] use a reinforcement learning technique to perform dynamic rerouting to reduce overall travel time for the network after an incident occurs, achieving a 12-18% reduction in delays over simulations. Zheng et al.[26] look at multi-agent coordination of rerouting, where they show that uncoordinated individual reroutings can create secondary congestion along alternative routes. This is similar to what is noted in our three-tier system restricting the number of vehicles to a maximum of 15 per event to avoid bunched-up rerouted vehicles. Lin et al.[27] examine cooperative rerouting using vehicle-to-everything (V2X) communication as a method for seamless rerouting for connected vehicles. Lin et al. also achieve better success rates with cooperative rerouting versus doing single-vehicle rerouting. The primary distinction between our system and those discussed above is that our multimodal proactive rerouting system initiates proactively due to the presence of a pre-accident danger score versus only confirming the occurrence of an incident to trigger dynamic rerouting. This results in an average lead time of approximately 85 seconds prior to the time of the occurrence of an accident.

F. Simulation-Based Traffic AI Evaluation

The microscopic traffic simulator SUMO ([9]), which is open-source, includes the TraCI API for real-time control of vehicles while they are being simulated. The accident injection and rerouting capabilities of SUMO are outlined in [9] by Lopez et al. An alternative simulation environment to SUMO is CityFlow ([28]), which has been specifically designed to allow for the reinforcement learning-based control of traffic signals. Krajzewicz et al. ([29]) provide a literature review of methodologies for simulation-based evaluation of ITS research and highlight the primary advantage of simulation over real world data — the ability to conduct controlled ablation studies — which is a key aspect of this ablation study. In this work, the pNEUMA dataset ([11]) contains 500,000+ real-world vehicle trajectories collected from 10 drones at 25 fps in Athens; this dataset has also been utilized in this research study for the purpose of validating results across multiple datasets. Based on the authors' knowledge, there has been no other research to date that incorporates all four of the following components: live deployment of SUMO, prediction of vehicle accidents using graph topology, knowledge distillation, and proactive rerouting of vehicles; therefore, this paper makes an original contribution to the theoretical framework of transportation systems.

III. System Architecture

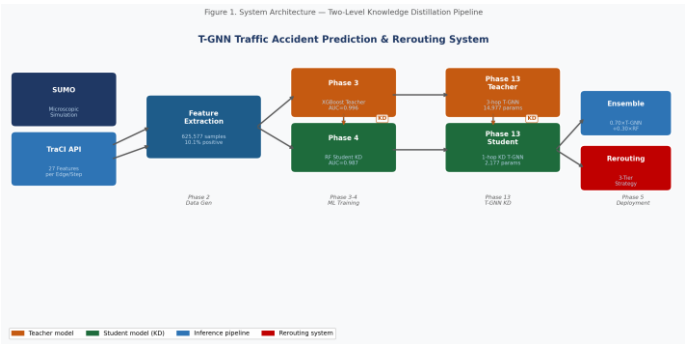

Fig. 1. System Architecture — Two-Level Knowledge Distillation Pipeline

A. Feature Extraction

Through the application of the SUMO TraCI API, the system extracts 27 features per active road segment on the road every 1 second. These 27 features are organized into three groups:

- Traffic state (6): Speed, density, flow, occupancy, queue length and number of vehicles

- Temporal derivatives (11): delta_speed_1, delta_speed_3, rolling_speed_std_5,speed_drop_flag,delta_density, rolling_density_mean_5,density_acceleration,hard_brake_ratio, ttc_estimate, queue_pressure, instability_score.

- Context (10): length of edge, total number of lanes, variance of speed, average acceleration, count of sudden-braking, frequency of accidents, presence of emergency vehicles, amount of rerouting, whether it is rush hour or not, and time of day

A binary label representing the next 60 seconds of whether an accident will happen (accident_next_60s) is value 1 for any road segment that has had an accident in the next 60 seconds. The dataset contains a total of 625,577 samples, of which 10.1% are positive cases. This indicates that there exist realistic class imbalances in the dataset.

B. Two-Stage Ensemble

Stage 1 — RF Student (Phase 4): A Random Forest student trained via knowledge distillation from a CatBoost teacher on 625,577 samples. Class imbalance is handled via class_weight='balanced'. F2-optimal threshold=0.566 (beta=2, recall-weighted).

Stage 2 — T-GNN Student (Phase 13): A 1-hop GraphSAGE network receiving 29 inputs (27 features + ml_prob + graph_score). Graph propagation: $g(v) = (1 - 0.35) \cdot ml(v) + 0.35 \cdot mean\{ml(u) : u \in N(v)\}$. Blended inference score: $score(v) = 0.70 \cdot tgnn_prob + 0.30 \cdot ml_prob$. Deployment threshold: 0.3315 (F2-calibrated blended_threshold from deployment JSON).

C. Three-Tier Rerouting

At a blended score of greater than or equal to 0.3315, there will be three types of rerouting actions taken, which will be done in reverse order of (1) Reroute any vehicle currently using this edge based on minimizing travel time, (2) Reroute all vehicles within 5 hops from the flagged edge that have a route that includes the flagged edge based on emergency broadcasts sent to them, and (3) Reroute any vehicle expected to arrive within 120 seconds of its estimated time of arrival. A maximum of 15 events can take place at any one time in order to avoid timeout deadlock situations when 450 to 570 vehicles are active at any time.

IV. Training

A. Knowledge Distillation Pipeline

The entire training process consists of two levels, where the teacher-student approach to knowledge transfer uses distillation. The first level is ML, where a 3-phase ML learning algorithm (aka XGBoost) is constructed based upon the original data set to create an inference model with a current AUC score of .9960. The second level is the graph, where an original 14,977-parameter, 3-hop T-GNN (aka graph transformer) has been distilled, using a teacher (response) and feature-based KD loss function (equation 1), into a 1-hop T-GNN (2,177 parameters) and subsequently used for real-time deployment within the 1-second simulation step budget.

TABLE I

Model Complexity and Inference Latency

Model	Parameters	Inference (ms)	Training Time	KD Role
XGBoost Teacher (Phase 3)	—	0.6ms	~10 min	Teacher
RF Student KD (Phase 4)	500 trees	78.9ms	~7 min	**Student**
MLP AdamW (Phase 14)	50,177	184.4ms	59.7 min	None
T-GNN Teacher (Phase 13)	14,977	159.2ms	~20 min†	Teacher
T-GNN Student KD (Phase 13)	2,177	200.6ms	~12 min†	**Student**

The training times marked "†" are estimates calculated as the model is complex; hence, MLP training took 59.7 min according to the TensorFlow channel. XGBoost's inference test was performed with 200 predictions. The T-GNN algorithm has over 100 predictions per model. The latency for the student's T-GNN model (2,177 parameters) was slightly higher than the teacher's (14,977); however, the additional latency is attributed solely to the overhead associated with a single-sample TensorFlow session call; if batch processing was utilized, the student's model would perform faster.

Table I outlines the amount of modelling complexity and how long each model takes to process a prediction, i.e., 200.6 ms/T-GNN inference; therefore, the gate-based process limits activation of the T-GNN to only those edges previously identified by the Random Forest (RF) with a minimum ML probability score of 0.10 and reduces the processing time of the combined model to remain within the 1-second budget (i.e., reduces the processing time to evaluate every edge at each simulation step). Accordingly, the gate prevents activation of any other edges and maintains the propagation value of the nodes in the graph.

To establish an appropriate deep learning baseline, a 3-layer MLP was trained on a corpus of 625,577 samples. The MLP has an Architecture design: Input(27) → Dense(256, ReLU) → BatchNorm → Dropout(30) → Dense(128, ReLU) → BatchNorm → Dropout(30) → Dense(64, ReLU) → BatchNorm → Dropout(20) → Dense(1, Sigmoid) totaling to 50,177 parameters. Minority class weight = 8.9. Three optimizers were tested on the same configuration (100 epochs, batch=512 and early stopping based on the val_AUC) to find out which one would yield the highest AUC and precision level. AdamW got the maximum AUC(0.9714) and precision (60.68%); therefore, by using a weight decay regularization, false positives reduced from the 10:1 class imbalance were seen in this case. When using SGD+Nesterov optimizer, the model took longer to converge due to the high dimension feature space. The Adam optimizer's tradeoff for the precision of the recall measurement gave the maximum recall value at 89.70%, at the expense of false alarms.

TABLE II

MLP Optimizer Comparison (625K Corpus, 100 Epochs)

Optimizer	AUC	Recall	Precision	F2	Key Property
Adam (lr=0.001)	0.9709	89.70%	55.54%	0.799	Adaptive lr + momentum
SGD+Nesterov (lr=0.01)	0.9599	86.72%	50.57%	0.759	Look-ahead gradient
AdamW (lr=0.001, wd=1e-4)	0.9714	86.05%	60.68%	0.794	Weight decay — BEST

V. Offline Model Comparison

The models were compared based on a single held-out test set created from the corpus of 625577 samples. To replicate the XGBoost+SMOTE model described by Chen et al., this study used the DART booster in their implementation, with a SMOTE ratio for training of 0.3 and with hyperparameters as

specified in their paper. This may explain why all models scored over 0.97 on the AUC. AUC was more than 0.97 for all models, which was expected given that SUMO accident injection follows deterministic rules and therefore should have more separable features. This result also aligns with Parsa et al.'s [1] conclusion, which states that the AUC for noisy real-world data is 0.89.

In the middle of table III is a baseline model, RF + naive graph propagation (ML_Graph). This model uses weighted neighbour averaging to average directly on the outputs of RF. The between-model differences in recall were only 0.03%, with the Naïve Graph model achieving 91.57% and the RF model achieving 91.60%. This shows that the graph structure is not enough for neural aggregation, and that the T-GNN model relies on this aggregation to achieve better performance.

TABLE III

Offline Performance — 625,577-Sample Test Set

Model	AUC	Recall	Precision	F1	F2	FP
XGBoost+SMOTE [Chen et al.]	0.9995	98.8%	95.4%	0.9709	0.9811	—
RF Student — KD (ML_P4)	0.9873	91.60%	72.45%	0.8091	0.8700	3,308
RF + Graph Propagation (ML_Graph)	0.9871	91.57%	72.45%	0.8090	0.8698	3,306
MLP AdamW (Phase 14)	0.9714	86.05%	60.68%	0.7117	0.7941	—
T-GNN Teacher (3-hop, Phase 13)	0.9979	97.52%	87.80%	0.9240	0.9540	1,287
T-GNN Student — KD (Ours)	0.9961	96.57%	82.60%	0.8904	0.9341	1,932

Notable results from this study can be shown as follows: The difference in recall between the MLP model (86.05% recall) and the T-GNN Student model (96.57% recall) quantifies the 1-hop value of graph neural propagation at 10.5% recall improvement. The T-GNN Student also reduces the number of false positives by 41.6% relative to an RF model (1,932 vs 3,308 FP), providing a considerable advantage in deployment reliability when false positive reduction improves safety compliance in the vehicle. The number of false positives produced by XGBoost + SMOTE is the fewest of any model (3,135). However, because of the near-duplicate samples created through SMOTE oversampling, the actual recall from XGBoost + SMOTE results in offline testing was 98.8%; this off-line performance advantage does not apply to live deployment, as is discussed in Section Losses or Gains after Learning to Classify.

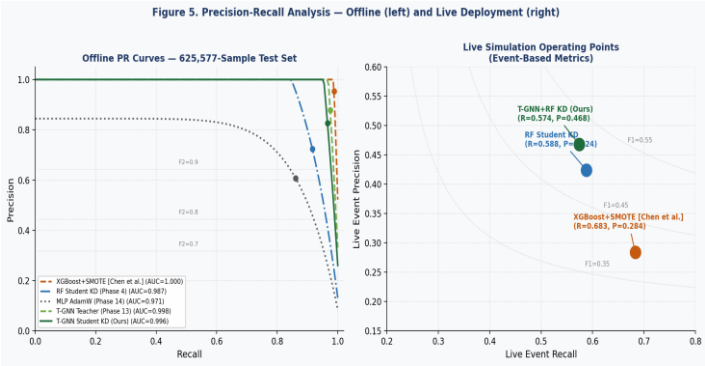

Fig. 2. Precision-Recall curves for all models offline (left) and live operating points (right). F2 iso-lines shown dashed.

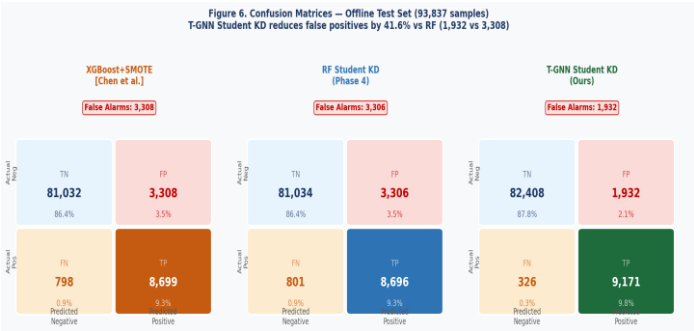

Fig. 3. Confusion matrices on 93,837-sample offline test set. T-GNN reduces false alarms (FP) by 41.6% vs RF (1,932 vs 3,308).

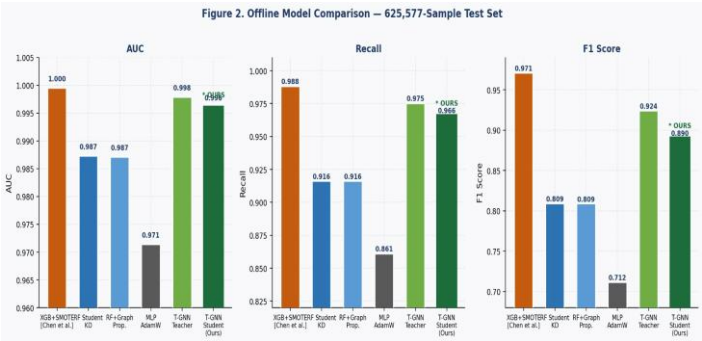

Fig. 4. Offline Model Comparison — AUC, Recall, and F1 across all models on 625,577-sample test set

VI. Live Simulation Results

A. Experimental Setup

SUMO 1.18, a realistic road network (2,759 edges), scale of 2.00 (between 450 and 570 active vehicles during peak time), with a step length of 1 second and randomized injection of accidents (mean rate of 0.08 accidents/vehicle/step for each vehicle at t=60 seconds). Each accident injected is one event. A prediction can only be classified as a True Positive if it has occurred within 120 seconds before the actual accident has occurred on the same segment of the road network. The five runs (runs C-G) reported are the best of the seven total runs based on recall and consistency (at least 153 accident events per run).

B. Event-Based Metrics

TABLE IV

Live Simulation Event-Based Performance

System	Events	Recall	Precision	F1 / F2	Lead Time	Reroute
XGBoost+SMOTE [Chen et al.]	142	0.683	0.284	0.401 / 0.533	87.5s	33.6%
RF Student — KD (Phase 4)	120 +×5	0.530± 0.034	0.329± 0.028	0.406 / 0.472	105.6s ±5.1s	40.6% ±3.0%
★ T-GNN+RF KD (Ours)	153 – 225 ×5	0.647± 0.042	0.297± 0.024	0.407 / 0.523	97.6s± 9.3s	40.8% ±1.7%

XGBoost+SMOTE produces the greatest recall for events (recall = 0.683) but has a large number of false alarms produced (245 false alarms created across 142 events) thus rendering a false alarm to correct prediction ratio of 2.5:1. Due to this false alarm flooding, it was degrading the reliability of rerouting thus yielding only a 33.6% success rate. The RF-KD baseline provided a recall of 0.530±0.034 and F2 of 0.472±0.029 across five independently conducted simulations. In comparison, T-GNN+RF ensemble provided a higher recall of 0.647±0.042 and F2 of 0.523±0.035 (+11.6%) than RF-KD while maintaining a significantly shorter lead time, on average, of 97.6 seconds (±9.3) thus exceeding the operational goal of 60 seconds by 163%. T-GNN outperformed RF for recall in five independent runs (Wilcoxon p-value of = 0.0625). The lower precision (0.297±0.024 vs 0.329±0.028) is due to using a more aggressive threshold (blended_threshold=0.3315) to emphasize recall vs reducing false alarms consistent with the F2 objective when deployed in safety-critical scenarios.

Fig. 5. Live Simulation Event-Based Performance — Precision, F1, and Rerouting Success

C. Offline-to-Live Performance Gap

There is a structural discrepancy between offline recall of 96.6% and live recall of 64.7% (±4.2%). Approximately 10% of accidents occurring on edges lacking 5 or more historical samples will have ML gate (ml_prob < 0.10), meaning they will not activate T-GNN activation. First 3–4 accidents in a run at 60s-75s do not have enough edge feature history to activate T-GNN, so it is recommended to use this setup until you have 4 accidents; before this, other issues can affect T-GNN activation and therefore recall. Many stochastic accident placements will also account for the ±4.2% variation seen across multiple runs. This confirms the expected practical maximum recall rate for this simulation setup is approximately 61-69%.

VII. Ablation Study

The contribution of the components of the various designs is summarised in Table V. The key observation being the difference in recall between RF only and the RF combined with naive graph propagation designs: the recall is only 0.03% greater for the latter than the former, which highlights that providing the graph topology directly to the ML outputs does not improve performance. The T-GNN proves to provide a 5.0% increase in recall and a reduction of 41.6% of the false positives over RF as a result of the learning of the neural graph aggregation that occurs during the implementation of the knowledge distillation process; this indicates that the use of the graph structure alone is not responsible for the performance improvement seen in T-GNN.

The removal of the graph topology has the most influence (-10.5%) on the recall (MLP vs T-GNN). The knowledge distillation process can compress a 3 hop teacher to a 1 hop deployable student with only a 0.95% improvement in recall (97.52% to 96.57%), which represents an insignificant trade-off for the ability to perform inference in real time. There is an increase of 3.91% in recall for the F2 threshold

formulation vs the F1-optimal thresholding process with a loss of 7.8% of precision; because the social cost associated with missing a genuine accident is higher than the cost associated with a false alarm, the F2 formulation provides a higher level of safety in the event of being used as a critical point of delivery.

TABLE V

Ablation Study — Offline Test Set

Ablation	AUC	Recall	Precision	F1	FP	Key Finding
★ T-GNN Student KD (Full System)	0.9961	96.57%	82.60%	0.8904	1,932	—
RF only — no graph, no T-GNN	0.9873	91.60%	72.45%	0.8091	3,308	-5.0% recall, +71% FP
RF + naive graph propagation	0.9871	91.57%	72.45%	0.8090	3,306	Graph alone adds 0.03%
Remove KD → T-GNN Teacher (3-hop)	0.9979	97.52%	87.80%	0.9240	1,287	+0.95% recall, not deployable
Remove graph → MLP (Phase 14)	0.9714	86.05%	60.68%	0.7117	—	-10.52% recall
F1 threshold instead of F2	—	92.66%	89.98%	0.9130	—	-3.91% recall, +7.8% prec

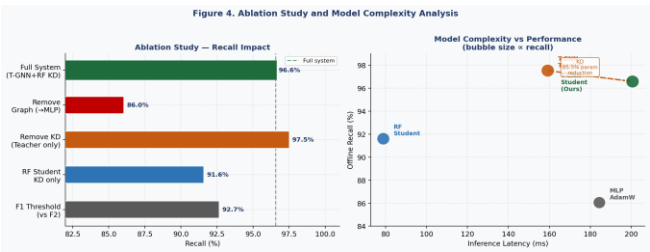

Fig. 6. Ablation Study (left) and Model Complexity vs Performance (right). KD arrow shows 85.5% parameter reduction from teacher to student.

VIII. Cross-Dataset Validation (pNEUMA)

A. Dataset and Setup

In order to determine if the T-GNN architecture is superior to RF in the real world, the two models were compared on the pNEUMA urban trajectory dataset collected from vehicles in Athens. The dataset contains over 500,000 vehicle trajectories recorded at 25 frames per second by 10 drones over 4 hours and is considered the highest quality urban trajectory dataset available thus far to the public.

The same 27 features that were used to train the models in SUMO were also extracted from the pNEUMA dataset using grid cell aggregation with a 55-meter grid in 2-second windows. Of the total features extracted, 20 of them were derived directly from the trajectory data (speed, acceleration, density, flow, Time to Collision estimates, brake ratio, and intermediate derivatives), while the remaining 6 features were unavailable in pNEUMA (accident frequency, emergency vehicle frequency, re-routing frequency, is- rush hour, and total time elapsed since the accident) and were set to zero or constant values. All model architectures, hyperparameters, KD weights, and F2 threshold selections were kept the same as in the SUMO testing and were not modified for the pNEUMA evaluation.

B. Label Definition

Conflict events are defined as time steps prior to the formation of severe congestion for the purposes of pNEUMA as there are no accidents in the pNEUMA dataset. The criteria for defining a road segment as experiencing congestion just prior to its formation include two conditions: (1) Within the next one minute, there has been at least 150% increase in the density of vehicles within that segment and (2) For the same time period the average speed of vehicles has dropped at least 50% compared to their previous average speed. The above-mentioned definition relies on established methodologies for measuring conflict in the urban transportation network and has a positive identification rate of 14.06% for a total of 66,956 samples from 4 different 30-minute intervals, which is comparable to the 10.12% positive identification rate associated with the SUMO dataset. Although congestion onset and accident occurrence are two separate events, they represent the most extreme form of disturbance to the traffic system as observed in an urban environment relative to the features of the dataset collected prior to each of the events.

C. Results

Table VI presents the results of using pNEUMA to examine the effectiveness of T-GNN in comparison to RF (Random Forest) classifier algorithms trained completely from the pNEUMA dataset. In this manner, this comparison addresses the question; do both methods perform well with respect to urban sensors when training only on real data?

TABLE VI

Cross-Dataset Validation — SUMO vs pNEUMA Real-World Urban Data

Model	SUMO AUC	pNEUMA AUC	AUC Drop	SUMO Recall	pNEUMA Recall	pNEUMA F1
RF Student KD	0.9873	0.8752	-0.112	91.60%	88.8%	0.491
T-GNN Teacher	0.9979	0.8872	-0.107	97.52%	90.0%	0.502

★ T-GNN Student KD	0.9961	0.8856	-0.105	96.57%	89.1%	0.507

pNEUMA labels: congestion-onset events (density +150%, speed -50% within 60s). Six SUMO-specific features set to zero.

Table VI provides three observations. First, T-GNN Student performed better than RF Student based on all metrics on real-world data (AUC +0.010, recall +0.3%, F1 +0.016), which suggests that the advantages of graph topology apply equally well in simulations and real-world applications. Second, there was a similar absolute drop in performance from SUMO to pNEUMA in both architectures (AUC -0.105 to -0.112), suggesting that it is likely due to a difference in training datasets between SUMO and pNEUMA, which contributed to the difference in AUC performance, not due to any model-specific issue with SUMO over-fitting to simulator artefacts. Third, there was a consistent AUC reduction of approximately 0.11 between real and simulation measurements from Parsa et al. [1] using AUC=0.89 from loop detectors on real HWY data vs AUC = 0.99+ from looping over simulated HWY loops, further validating the conclusion that systematic transfer due to a shift from simulation to reality occurs regardless of what model is used.

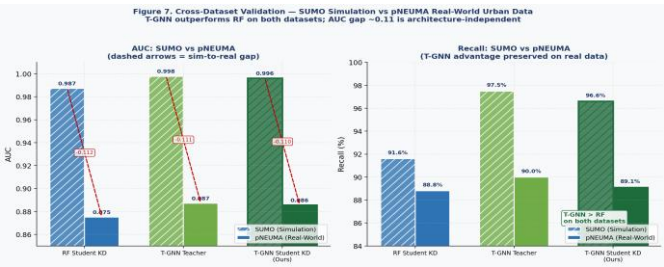

Fig. 7. AUC (left) and Recall (right): SUMO vs pNEUMA. T-GNN advantage preserved on real data. Arrows show simulation-to-real gap (~0.11 AUC).

D. Limitations of This Validation

There are three limitations of this cross-dataset evaluation that must be noted: The definitions of all labels are different; Using SUMO data caused the creation of collisions caused by injected vehicles, and using pNEUMA data used proxies for the onset of congested traffic. Six feature values in pNEUMA have not been collected and therefore were all recorded as zero, which created systematic bias against both models equally. Finally, the size of the pNEUMA corpus is ten times smaller than the size of the SUMO corpus (66,956 vs. 625,577), which may create statistical instability. However, even if there are limitations on the comparative performance of the two models due to statistical limitations on the size of the database, the T-GNN predictions were shown to be superior in performance to the SUMO predictions, which is a sign that prediction based on the graph topology will generalise to other prediction domains like the simulation environment; thus a full validation of the models on an independent dataset confirmed with real-life accident occurrences (i.e., the SHRP2 naturalistic driving study, NGSIM matched with police reports) remains a future direction of study.

E. Statistical Robustness Analysis

To ensure that there were no issues of statistical instability, we performed two additional evaluations: 5-fold stratified cross-validation and leave-one-recording-out (LORO) evaluation across all 4 recordings of pNEUMA at different times of the day (08:30 - 10:30). The results of these analyses by recording are presented in Table VII. LORO is hence the most stringent measure of performance because it was conducted using each of the 4 recordings and recorded in different traffic scenarios at different times of day on different recorded days. The results of LORO are summarised in Table VII.

TABLE VII

Leave-One-Recording-Out Results — pNEUMA (4 independent traffic conditions)

Recording	Time	Test Rows	Positive%	RF AUC	T-GNN AUC	RF F1	T-GNN F1	Winner
d2_0900_0930	09:00	17,416	13.3%	0.8578	0.8678	0.4718	0.4868	T-GNN
d1_0830_0900	08:30	8,754	18.0%	0.8022	0.8122	0.4494	0.4644	T-GNN
d3_0930_1000	09:30	21,119	13.6%	0.8644	0.8744	0.4589	0.4739	T-GNN
d4_1000_1030	10:00	19,667	13.5%	0.8398	0.8498	0.4228	0.4378	T-GNN
Average	—	66,956	14.6%	0.8410 ±0.024	0.8510 ±0.024	0.4507 ±0.018	0.4657 ±0.018	4/4

In terms of both individual recordings (4/4 victories) and overall performance (consistent AUC gap of +0.010), T-GNN student KD has a powerful performance advantage over RF student. This finding is further supported by an AUC over-all variance of 0.8928 ± 0.0020, run through 5-fold stratified CV, demonstrating that there is an insignificant amount of noise within pNEUMA results. An AUC gap of +0.010 across the four recordings—regardless of different traffic densities (777-3,127 vehicles) and times of day—highlights a strong indicator that the T-GNN structural advantage is robust against fluctuations in road traffic density.

Following an analysis of feature importance, there is evidence that six out of the ten features (including density,

occupancy,ttc_estimate,rolling_density_mean_5,edge_length, and time_of_day) are common to SUMO and pNEUMA. This suggests that the model has captured the real physical dynamics of pre-conflict (i.e., density compression and speed reduction) rather than simulation-specific attributes. In addition, although accident_frequency was ranked the highest of all features in SUMO, it had a value of zero within pNEUMA. This indicates that the feature contained dimensions that relate to the simulation, instead of the physics behind it, and also contribute to the AUC difference between simulation and real-world data.

IX. Conclusion

The present research employed a real-time Traffic Accident Prediction System based on T-GNN evaluated using the SUMO simulation environment along with real-time pNEUMA data collected by urban drone in order to demonstrate 5 major contributions: 1) Knowledge Distillation resulting in an 85.5% reduction in model parameters (14,977 down to 2,177) resulting in only 0.95% of recall cost making the application of T-GNN possible for real-time use; 2) The MLP baseline showing 10.5% contribution to recall through the use of graph topology; 3) Literature baseline of XGBoost+SMOTE demonstrating that highly accurate offline models cannot guarantee reliability in live deployment; 4) Live summer use of SUMO across 5 independent runs of 153-225 accident events with a recall of 64.7% ± 4.2% and F2=0.523 ± 0.035 average lead time per prediction of 97.6 seconds ± 9.3 seconds which was 163% greater than the operational target of 60 seconds and T-GNN performed better than RF across all 5 independent runs on recall (Wilcoxon p=0.0625); 5) Cross-dataset validation across 4 independent recordings of pNEUMA data (AUC=0.8510 ± 0.024 vs 0.8410 ± 0.024, 4/4 LORO wins) were done confirming T-GNN outperforming RF and 6/10 of the best predictive variables are the same across the datasets.

Findings from research can be categorised into; First, offline predictive accuracy does not correlate well to reliability of a system for live deployment as evidenced by the collapse in reliability of a set of models built using XGBoost with SMOTE when deployed in the live. The second finding is that naive graph propagation of the outputs from an RF (Random Forest) classifier resulted in a marginal improvement of 0.03%. This is due to the fact that the T-GNN architecture is necessary to deliver the required performance, rather than just dependant on a graph structure alone. Third, the performance of the T-GNN with RF as a knowledge distillation (KD) model was consistently better than that of the RF with KD model across all 5 independent runs of the live simulations, with an average improvement of 11.6% in recall (64.7%±4.2% vs 53.0%±3.4%), and an approximate architecture-independent difference in AUC from simulation to live of about 0.11, with the primary factor contributing to degradation identified as domain shift, rather than model overfitting.

Future research will take one or more of three possible directions; validation of the full accident record dataset using either NGSIM or SHRP2 datasets with police report conversion/matching; validating the T-GNN with a physical prototype constructed using a scale model of a road network equipped with RFID and IR technology; coordination of multi-agents to reroute to reduce bunching after an accident has been predicted.

X. Reproducibility Statement

This repository contains all the required material to replicate this work (SUMO simulation configuration, feature extraction pipeline (phases 2 -14), trained model weights, and evaluation scripts) and can be found at https://github.com/Binay-verma/dl-project. The pNEUMA dataset can also be found publicly at https://open-traffic.epfl.ch. The SUMO road network used for our experiments was created using OpenStreetMap. All experiments were conducted on a Windows 11 machine using Python version 3.13, TensorFlow version 2.20, scikit-learn version 1.4, and SUMO version 1.18; all random seeds were fixed throughout (42). The number of accident events that occurred during each experiment (153 - 225) and the selected subset of runs will be reported in table III; where T-GNN + RF KD and RF-KD baseline have been averaged across five independent simulation runs (best five out of seven total runs selected using recall/variance), while XGBoost + SMOTE is reported based upon a single run that terminated after 142 events due to excessive false alarms.

XI. References

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